



Bio-inspired Adaptive Locomotion via Torque-based Learning

A Case Study of SATA

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Command



Torque



Motion

1 Locomotion Is a Control Problem

THE PROBLEM

Position-based locomotion is accurate under known conditions but rigid — it struggles with compliance, disturbance handling, unknown terrain, and the sim-to-real gap.

→ Torque-based control may enable more compliant interaction and improve adaptation capability.

POSITION CONTROL

Command position



Rigid behavior



Poor adaptation

TORQUE CONTROL

Command force



Compliant interaction



Adaptive response

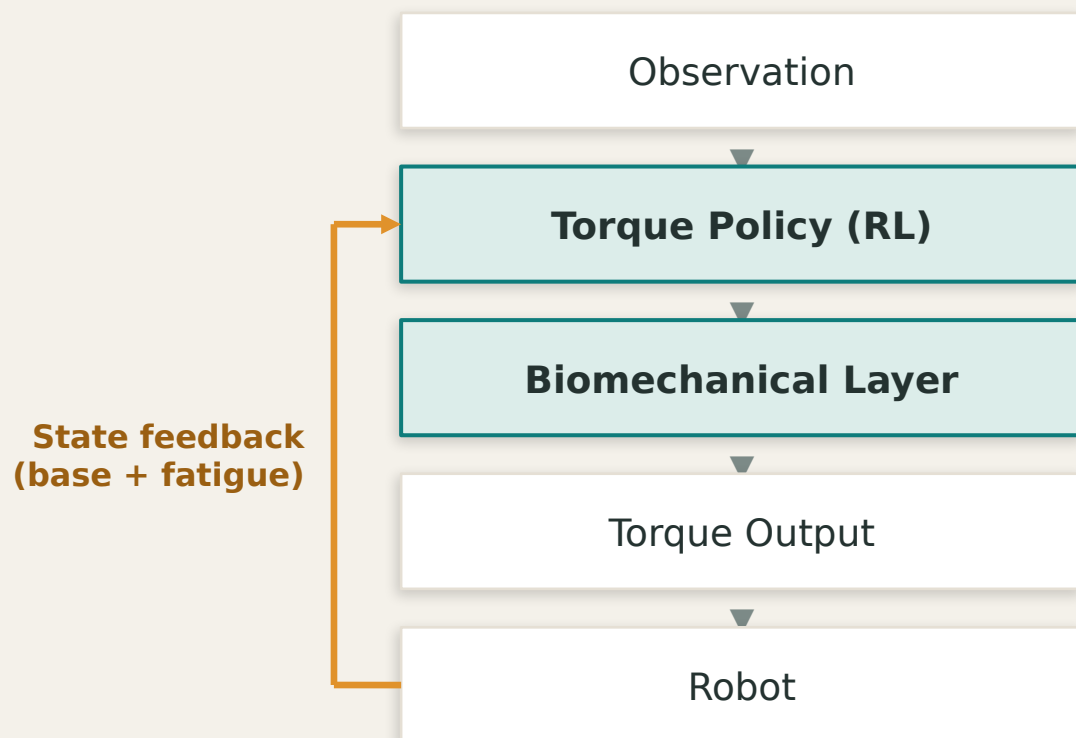
Adaptive behavior $\neq$ adaptive control

RESEARCH QUESTION

How can robots achieve safer and more adaptive locomotion in unknown environments?

2 SATA: Bio-inspired Adaptation in Torque Control

An RL torque policy wrapped by a biomechanical layer and a growth curriculum.



Biomechanical Model

activation · muscle · fatigue



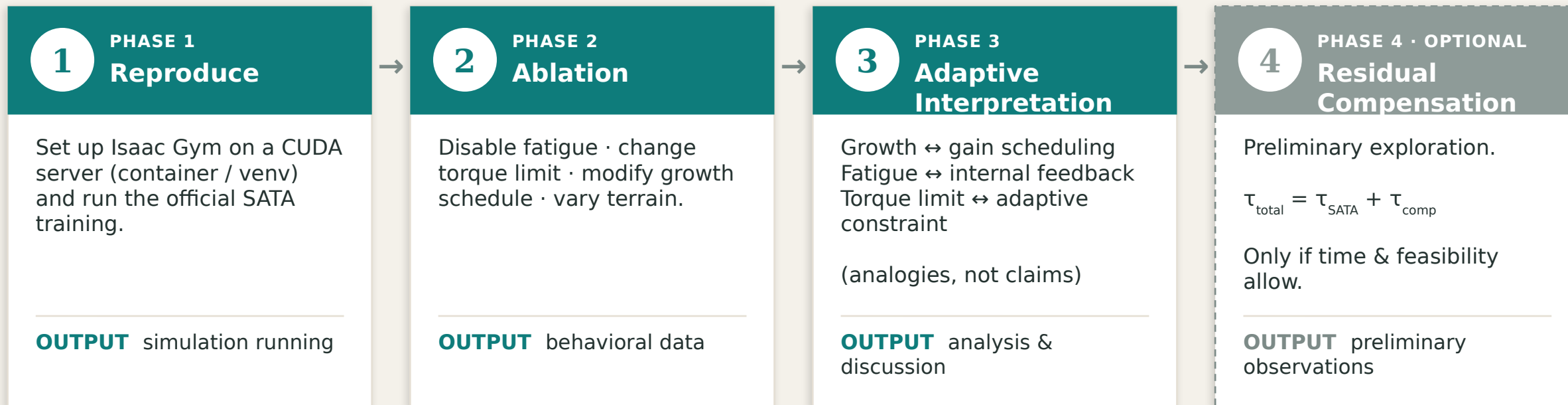
Growth Mechanism

torque limit · training curriculum · control frequency

Goal: smooth, safe torques that progressively unlock capability — demonstrated zero-shot deployment (sim-to-real).

3 Plan: Understand → Reproduce → Analyze

A feasible 3-phase study, with an optional extension — we interpret SATA, not redesign the RL.



APPROACH

We analyze adaptation instead of proposing new methods — we do not redesign the RL algorithm.

4 Expected Outputs & Contribution

We study adaptive behavior — not propose a new RL algorithm.

EXPECTED OUTPUTS

- 1 Simulation reproduction
- 2 Behavior analysis (ablation)
- 3 Adaptive control interpretation
- 4 Literature-grounded discussion

RESEARCH QUESTIONS

RQ1

Why torque control?

RQ2

What creates adaptation?

RQ3

How does it relate to adaptive control?

RQ4

Can lightweight compensation help?

IN ONE LINE

This project studies how bio-inspired torque control produces adaptive locomotion, and interprets it through adaptive and robust control perspectives.